

# CariSurg MedTech Pathways Research Project

## Preliminary Proposal

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**Project Title:** AI-Assisted Emergency Department Triage Pilot at Mercer General Hospital

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### Statement of the Problem:

Triage nurses in Mercer General Hospital's Emergency Department must rapidly assess patients during periods of high demand, increasing the risk of inconsistent prioritisation and delayed identification of high-acuity cases. Although recent studies suggest AI-assisted triage can improve patient prioritisation, most systems have not been validated using local emergency department data. This project proposes a 12-week pilot using de-identified Mercer ED records to evaluate whether AI-assisted acuity predictions align with current triage decisions and improve consistency in patient prioritisation.

### Previous Work:

The study is titled **“Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review.”** Emergency departments need faster and more accurate ways to triage patients. The authors reviewed studies from EMBASE, Ovid MEDLINE, and Web of Science to examine how AI and machine learning support ED triage. They found that machine learning improves triage prediction, disease identification, risk assessment, hospital admission prediction, and resource planning. The authors also noted that hospitals need more real-world testing across different settings because many studies used limited or retrospective data.

The study **“AI-driven triage in emergency departments: A review of benefits, challenges, and future directions”** addresses overcrowding, limited resources, and inconsistent patient prioritisation in emergency departments. The authors reviewed peer-reviewed studies from 2015 to 2024 using PubMed, Scopus, IEEE Xplore, and Google Scholar. They found that AI-driven triage improves patient prioritisation, reduces waiting times, supports resource planning, and helps clinical staff manage high-pressure ED settings. The authors identified poor data quality, algorithmic bias, clinician trust, and ethical concerns as key barriers. They recommended stronger algorithms, better use of wearable technology, clinician training, and clear ethical guidelines before wider use in emergency care.

The study **“TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers”** focuses on emergency department overcrowding, limited resources, and the pressure on triage staff to prioritise patients quickly and accurately. The authors built and tested several AI models, including SVM, Random Forest, ANN, GBM, Linear Regression, XGBoost, and a stacking model, to predict patient triage levels using the Korean Triage and Acuity Scale. The stacking model produced the strongest result, with 80.05% accuracy, while SVM and GBM also performed well. The authors identified the single dataset as the main limitation because it reduces the model's use across hospitals with different patients, seasons, and clinical conditions.

The study “**A Human-Centered Hybrid AI Framework for Optimizing Emergency Triage in Resource-Constrained Settings**” addresses the need for safer, faster emergency triage in healthcare facilities with limited staff, equipment, and clinical resources. The authors developed a human-centred AI system that uses patient vital signs, available resources, and clinician workload while allowing clinicians to review and override recommendations. The system achieved 90% accuracy, reduced missed high-acuity cases, shortened clinician decision-making time, and improved critical resource use. The authors identified the need for broader real-world validation and ongoing ethical and clinical supervision before full use.

The study “**Application of artificial intelligence in triage in emergencies and disasters: a systematic review**” addresses the difficulty of triaging large numbers of injured patients during disasters and emergencies, where traditional manual methods slow decisions, limit patient tracking, and weaken resource allocation. The authors conducted a systematic review following PRISMA guidelines, searching PubMed, Scopus, Web of Science, Google Scholar, and article references up to September 2024. They reviewed 19 high-quality studies and found that AI and smart triage systems improved patient care through faster triage, real-time vital sign monitoring, better data sharing, and stronger resource management. The authors identified major limitations, including varied study designs, heavy use of simulated settings, English-only evidence, trust concerns, staff training needs, equipment shortages, and data privacy risks.

### **Gap 1: Local validation against Mercer General’s current triage decisions**

Several studies report strong AI triage performance. Tyler found that AI/ML improved triage prediction and risk assessment. Araouchi and Adda tested models for KTAS triage level prediction and reported about 80% accuracy. Bhuvaneshwari reported improved detection of high-acuity cases. The studies repeatedly leave uncertainty about whether these tools perform well in a different hospital setting. Araouchi and Adda used a single dataset. Tyler noted that many studies rely on limited or retrospective evidence. Bhuvaneshwari et al. called for broader real-world validation. Mercer General cannot assume that a tool tested elsewhere will safely identify high-acuity patients in its own ED. Local patient mix, staffing patterns, documentation habits, and triage behaviour can all affect performance. If the tool misses high-acuity patients, it could contribute to under-triage and delayed care. Mercer could run a 12-week retrospective pilot using de-identified ED triage records. The pilot would compare AI-assisted triage outputs with current triage decisions and later clinical outcomes. The main outcome could be whether the tool improves identification of high-acuity patients compared with current triage decisions. This is feasible because it uses existing data and does not change live patient care.

### **Gap 2: Clinician trust and usability before live deployment**

Da’Costa identified clinician trust, ethical concerns, and training needs as major barriers to AI triage adoption. Tahernejad et al. also found trust, staff training, equipment, privacy, and implementation challenges. Bhuvaneshwari included clinician override and feedback, showing the importance of keeping clinicians involved. Although many studies focus on accuracy, the literature provides less evidence on whether ED clinicians understand, trust, and would realistically use AI recommendations during triage. This matters because even an accurate tool will fail if triage nurses do not trust it, understand it, or find it useful during busy shifts. Poor trust can lead to ignored alerts, over-reliance, or unsafe use. ED leaders need to know whether the tool supports clinical judgement rather than disrupting it. Mercer could run a 12-week shadow-mode pilot where triage nurses review AI-assisted recommendations after normal triage

decisions have been made. Nurses could rate whether the recommendation was clear, useful, and clinically reasonable. The main outcome could be clinician agreement and perceived usefulness of AI-assisted triage recommendations.

### **Proposed Solution:**

This project aims to build a small AI-assisted triage evaluation prototype for Mercer General Hospital's Emergency Department. The prototype will use de-identified past ED records. It will compare the estimated acuity level with existing triage decisions to assess consistency in patient prioritisation. The falsifiable target is that the prototype matches current triage decisions in at least 75% of reviewed cases and identifies high-acuity cases with at least 80% sensitivity. The project will include one simple prediction model, a basic results dashboard, and a final evaluation report.

### **References :**

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